

Topic 19: Organic Nitrogen Compounds: Amines, Amides, Amino Acids and Proteins

Students will be assessed on their ability to:

19.1	understand the nomenclature of amides, amines and amino acids and be able to draw their structural, displayed and skeletal formulae
19.2	understand the reactions of primary aliphatic amines (using butylamine as an example) and aromatic amines (using phenylamine as an example) with: - i water to form an alkaline solution - ii acids to form salts - iii halogenoalkanes - iv ethanoyl chloride - v copper(II) ions to form a complex ion
19.3	understand that amines are miscible with water as a result of hydrogen bonding, and the reasons for the difference in basicity between ammonia, primary aliphatic amines and primary aromatic amines
19.4	understand, in terms of reagents and general reaction conditions, the preparation of primary aliphatic amines: - i from halogenoalkanes - ii by the reduction of nitriles
19.5	know the preparation of aromatic amines by the reduction of aromatic nitro-compounds using tin and concentrated hydrochloric acid
19.6	be able to describe the reaction of aromatic amines with nitrous acid to form benzenediazonium ions, followed by a coupling reaction with phenol to form a dye
19.7	understand that amides can be prepared from acyl chlorides
19.8	be able to describe: - i condensation polymerisation for the formation of polyamides such as nylon and proteins - ii addition polymerisation, including poly(propenamide) and poly(ethenol)
19.9	be able to draw the structural formulae of the repeat units of the polymers in 19.8
19.10	be able to comment on the physical properties of polyamides and the solubility in water of the addition polymer poly(ethenol) in terms of hydrogen bonding, including soluble laundry bags or liquid-detergent capsules (liquid tabs)
19.11	be able to describe experiments to investigate the characteristic behaviour of amino acids limited to: - i acidity and basicity and the formation of zwitterions - ii effect of aqueous solutions on plane-polarised monochromatic light - iii formation of peptide bonds by condensation polymerisation

	Further suggested practicals: - i carry out some of the reactions of amines from 19.2 i, ii and v - ii prepare an azo dye from 19.6 - iii carry out reactions of amino acids from 19.11 i - iv prepare nylon-6.6 or 6.10 from 19.8
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